K.A.W.H.I – Kinetic Adaptation via Wasserstein Heuristics & Identity

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Data

- **SportVu** Tracking Data + **PBP** Shot Data
- 2015-16 NBA Season
- 631 Games
- 86957 Shots

Method

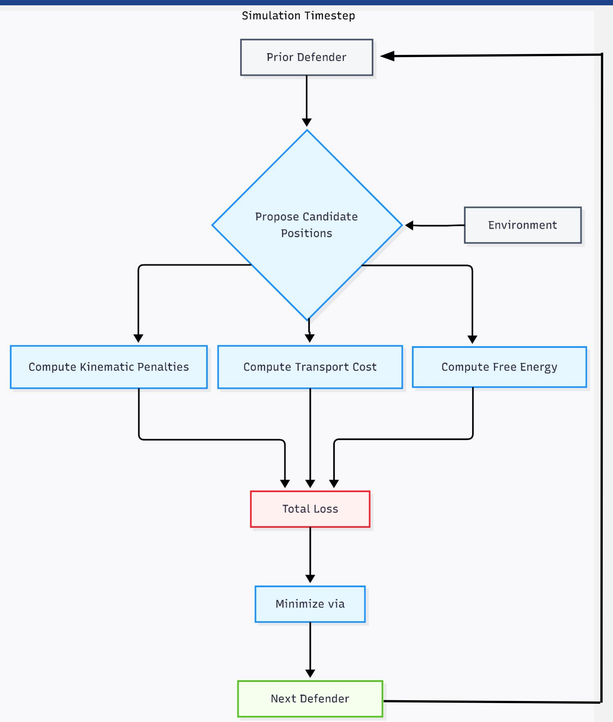
- **The JKO scheme:**
  Minimizes ↓ for the defense

$$\underset{y}{\operatorname{argmin}} \left(\mathcal{E}(y) + \frac{1}{2\tau} W_2^2(y, x_t) \right)$$

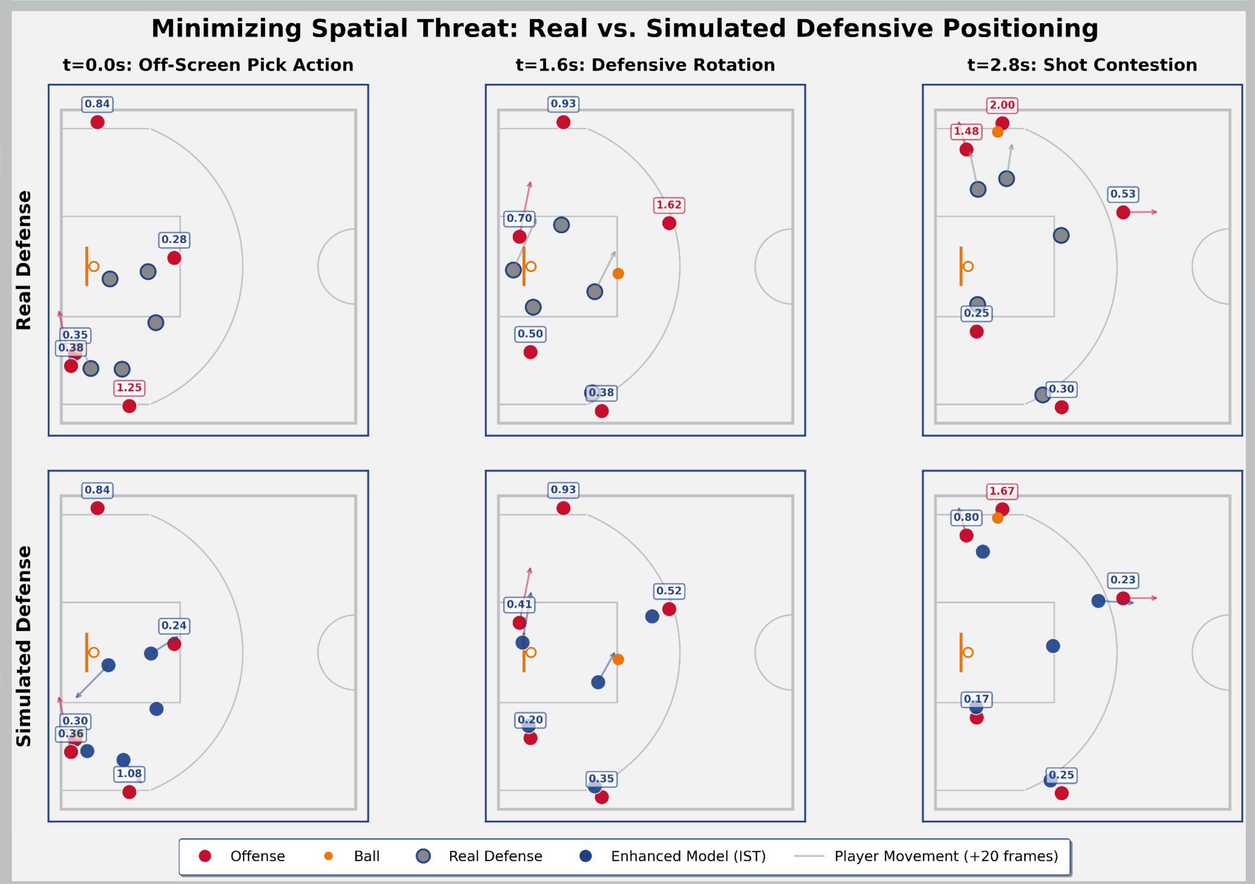
Potential 'Threat' — Distance To the Next Formation

- **Bayesian Optimization**: automated Hyperparameter Tuning using **Optuna**.
- **JAX**: Python library for high speed array computation and automated differentiation

Simulation Loop

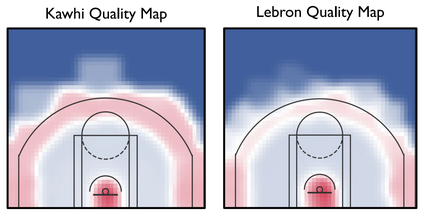


Can the transition of a basketball defense be mathematically modeled as a Wasserstein Gradient Flow?



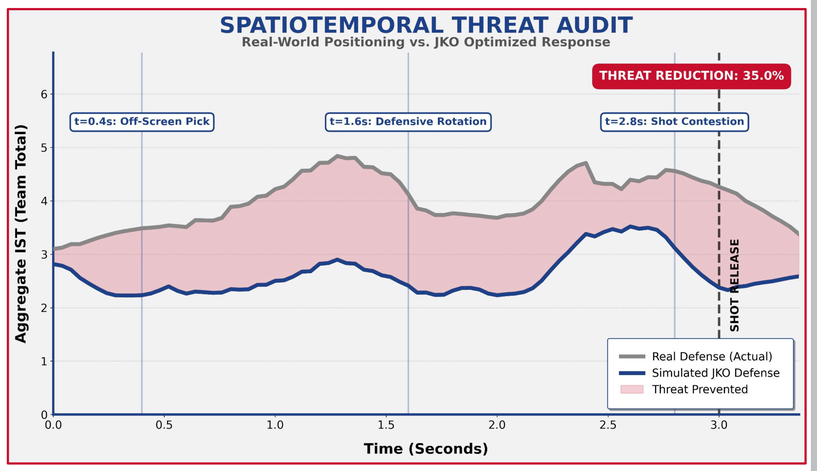
Instantaneous Shot Threat (IST) = Q · B · O

Shooter Quality (Q) — Expected Points Per Shot based on player coordinates

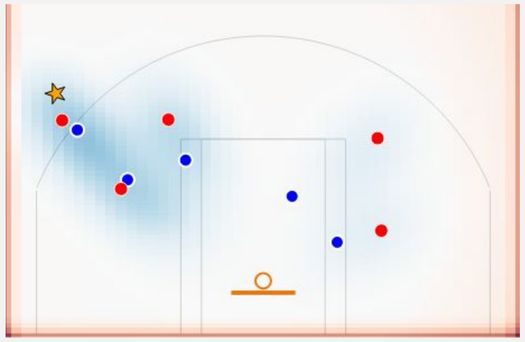


Ball Proximity (B) — Inverse distance between shooter and ball

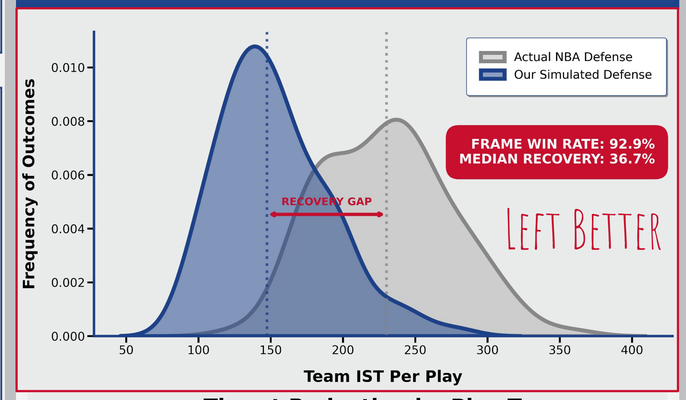
Openness Factor (O) — Distance to the nearest defender

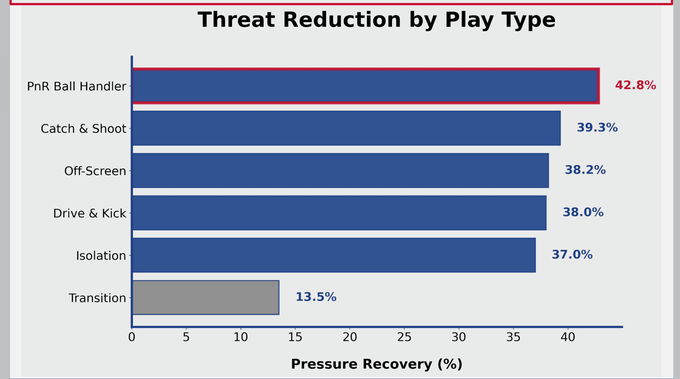


Potentials



How Good Is Our Model?





Limitations/Future Work

- Find a better way to control defender speed
- Add more realistic physical constraints
- Make model more sensitive to offensive plays
- Isolate individual player defensive contributions
- Engineer a more robust evaluation metric